

MATH 8090: Estimating Trend and Seasonality

Readings: CC08 Chapter 3; SS17 Chapter 2; BD16 Chapter 1.5

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Week 2

Agenda

The Classical Decomposition Model

Trend Estimation

Trend Removal through Differencing

Estimating Seasonality

The Classical (Additive) Decomposition Model

- ▶ The additive model for a time series $\{Y_t\}$ is

$$Y_t = \mu_t + s_t + \eta_t,$$

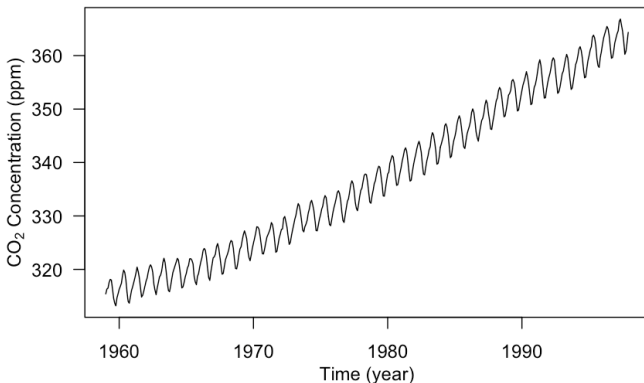
where

- ▶ μ_t is the **trend** component
 - ▶ s_t is the **seasonal** component
 - ▶ η_t is the **random (noise)** component with $\mathbb{E}(\eta_t) = 0$
- ▶ Standard procedure:
 1. Estimate or remove the trend and seasonal components.
 2. Analyze the remainder, $\hat{\eta}_t = y_t - \hat{\mu}_t - \hat{s}_t$.
 - ▶ This week, we focus on the first step

Mauna Loa Atmospheric CO₂ Concentration Revisited

Monthly atmospheric concentrations of CO₂ at the Mauna Loa Observatory [Source: [Keeling & Whorf, Scripps Institution of Oceanography](#)]

```
```{r}
data(co2)
par(mar = c(3.8, 4, 0.8, 0.6))
plot(co2, las = 1, xlab = "", ylab = "")
mtext("Time (year)", side = 1, line = 2)
mtext(expression(paste("CO"[2], " Concentration (ppm)")), side = 2, line = 2.5)
```
```



Estimating Trend for Nonseasonal Model

Start with a **trend** + **noise** model.

- ▶ Assuming $s_t = 0$ (i.e., there is no “seasonal” variation), we have

$$Y_t = \mu_t + \eta_t,$$

with $\mathbb{E}(\eta_t) = 0$

- ▶ Methods for **estimating trends**
 - ▶ **Parametric approach**: Least squares regression
 - ▶ **Nonparametric approach**: Smoothing
- ▶ Alternatively, one can **remove trend** by **differencing time series**

Trend Estimation: Linear Regression

- ▶ The additive nonseasonal time series model for $\{Y_t\}$ is

$$Y_t = \mu_t + \eta_t,$$

where the trend is assumed to be a linear combination of known covariate series $\{x_{it}\}_{i=1}^p$

$$\mu_t = \beta_0 + \sum_{i=1}^p \beta_i x_{it}.$$

- ▶ Here we want to **estimate** $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ from the data $\{y_t, \{x_{it}\}_{i=1}^p\}_{t=1}^T$
- ▶ This is the familiar multiple linear regression framework.

Some Examples of Covariate Series $\{x_{it}\}$

► Simple linear regression model:

$$\mu_t = \beta_0 + \beta_1 x_t,$$

For example, x_t could be the carbon dioxide level used to explain temperature at time t .

► Polynomial regression model:

$$\mu_t = \beta_0 + \sum_{i=1}^p \beta_i t^i$$

► Change point model:

$$\mu_t = \begin{cases} \beta_0 & \text{if } t < t^*; \\ \beta_0 + \beta_1 & \text{if } t \geq t^*. \end{cases}$$

Parameter Estimation: Ordinary Least Squares

- ▶ Estimate the parameters using **ordinary least squares (OLS)**.

Question: What assumptions are required for OLS estimation and inference?

- ▶ Specifically, we minimize the following objective function:

$$\ell_{ols} = \sum_{t=1}^T (y_t - \beta_0 - \sum_{k=1}^p x_{kt} \beta_k)^2.$$

- ▶ The estimates $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ minimizing the above objective function are called the **OLS estimates of β** $\Rightarrow$ they are easiest to express in **matrix form**

The Model and Parameter Estimates in Matrix Form

- ▶ Matrix representation:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta},$$

$$\text{where } \mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_T \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{T1} & x_{T2} & \cdots & x_{Tp} \end{bmatrix}, \text{ and}$$

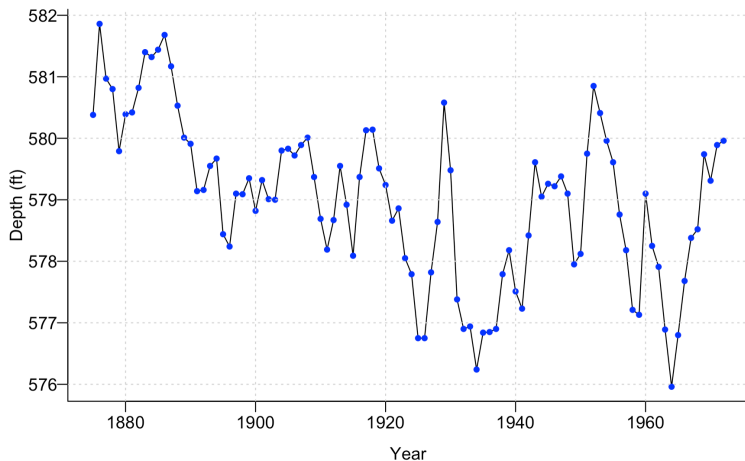
$$\boldsymbol{\eta} = \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_T \end{bmatrix}$$

- ▶ Assuming $\mathbf{X}^T \mathbf{X}$ is **invertible**, the OLS estimate of $\boldsymbol{\beta}$ can be shown to be

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y},$$

and the `lm` function in R calculates OLS estimates

Lake Huron Example Revisited



Let's **assume** there is a **linear trend in time** $\Rightarrow$ we need to estimate the **intercept** β_0 and **slope** β_1

The R Output

Call:

```
lm(formula = LakeHuron ~ yr)
```

Residuals:

| | Min | 1Q | Median | 3Q | Max |
|--|----------|----------|---------|---------|---------|
| | -2.50997 | -0.72726 | 0.00083 | 0.74402 | 2.53565 |

Coefficients:

| | Estimate | Std. Error | t value | Pr(> t) | |
|-------------|------------|------------|---------|----------|-----|
| (Intercept) | 625.554918 | 7.764293 | 80.568 | < 2e-16 | *** |
| yr | -0.024201 | 0.004036 | -5.996 | 3.55e-08 | *** |

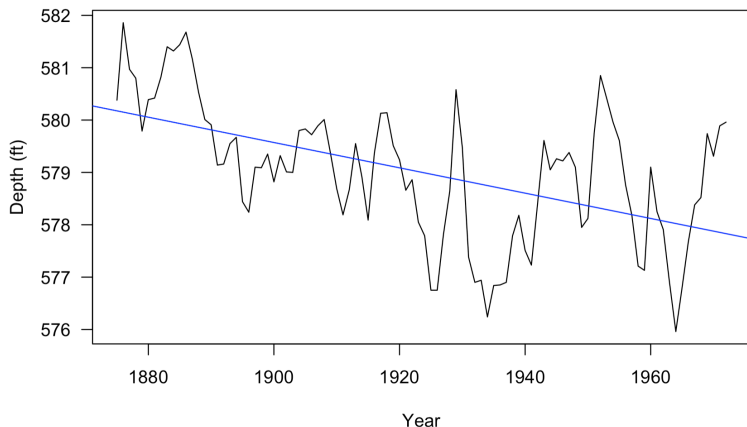
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Residual standard error: 1.13 on 96 degrees of freedom

Multiple R-squared: 0.2725, Adjusted R-squared: 0.2649

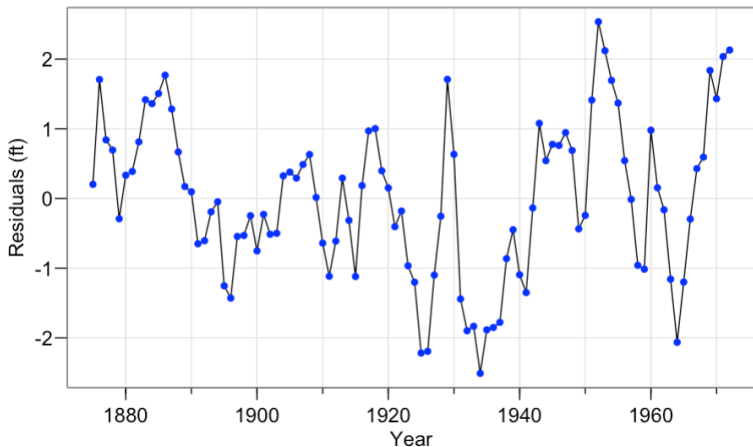
F-statistic: 35.95 on 1 and 96 DF, p-value: 3.545e-08

Plot the (Estimated) Trend $\hat{\mu}_t = \hat{\beta}_0 + \hat{\beta}_1 t$



$\hat{\beta}_1 = -0.0242 \text{ (ft/yr)} \Rightarrow$ there seems to be a decreasing trend

Plot the Residuals $\{\hat{\eta}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 t\}$



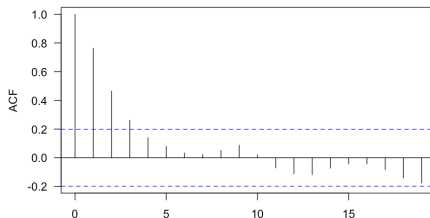
$\{\hat{\eta}_t\}$ seems to exhibit some temporal dependence structure, should we worry about the results we have (recall OLS makes an i.i.d. assumption)?

Statistical Properties of $\hat{\beta}_{OLS}$ with Correlated Errors

- ▶ Assume the components of $\mathbf{X}$ are not random, the OLS estimates $\hat{\beta}_{OLS}$ are **unbiased** for β

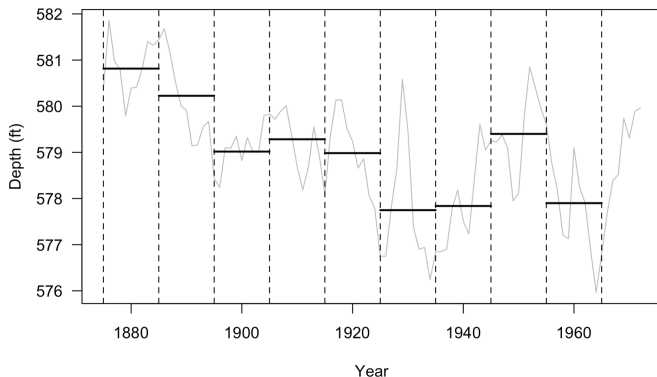
Proof:

- ▶ Since $\{\eta_t\}$ is typically not an i.i.d. process (see the acf plot below), statistical inferences regarding β will be invalid



Smoothing or Local Averaging

In certain situations, we may want to relax the “parametric” assumption on the trend $\Rightarrow$ a “nonparametric” approach. Here, we break the time series up into “small” blocks (each with 10 years of data) and average each block

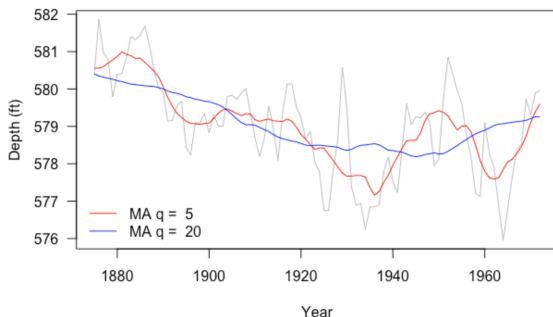


Doing this gives a very rough estimate of the trend. **Can we do better?**

Moving Average Smoother

A **moving average smoother** estimates the trend at time t by averaging the current observation and the q nearest observations from either side. That is

$$\hat{\mu}_t = \frac{1}{2q+1} \sum_{j=-q}^q y_{t-j}$$



q is the “smoothing” parameter, which controls the smoothness of 16/44

Exponential Smoothing

- ▶ Let $\alpha \in [0, 1]$ be fixed and define

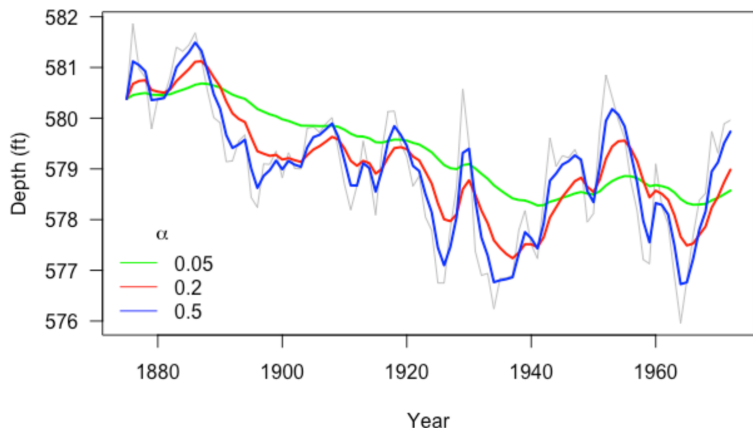
$$\hat{\mu}_t = \begin{cases} Y_1 & \text{if } t = 1; \\ \alpha Y_t + (1 - \alpha)\hat{\mu}_{t-1} & t = 2, \dots, T. \end{cases}$$

- ▶ For $t = 2, \dots, T$, we can rewrite $\hat{\mu}_t$ as

$$\sum_{j=0}^{t-2} \alpha(1 - \alpha)^j Y_{t-j} + (1 - \alpha)^{t-1} Y_1.$$

$\Rightarrow$ it is a one-sided moving-average filter with **exponentially decreasing weights**. The value of α controls the amount of smoothing.

α Is the Smoothing Parameter for Exponential Smoothing



The smaller the α , the smoother the resulting trend

Differencing

The final method we consider is **differencing**, which **removes** rather than **estimates** the trend

- ▶ We define the first order difference operator ∇ as

$$\nabla Y_t = Y_t - Y_{t-1} = (1 - B)Y_t,$$

where B is the **backshift operator** and is defined as $BY_t = Y_{t-1}$.

- ▶ For $q \geq 2$, define higher-order differences recursively by $\nabla^q Y_t = \nabla(\nabla^{q-1} Y_t)$.

- ▶ The backshift operator of power q is defined as $B^q Y_t = Y_{t-q}$

In next slide we will see an example regarding the relationship between ∇^q and B^q

Difference and Backshift Notation

The second order difference is given by

$$\nabla^2 Y_t = \nabla[\nabla Y_t]$$

Difference and Backshift Notation

The second order difference is given by

$$\begin{aligned}\nabla^2 Y_t &= \nabla[\nabla Y_t] \\ &= \nabla[Y_t - Y_{t-1}]\end{aligned}$$

Difference and Backshift Notation

The second order difference is given by

$$\begin{aligned}\nabla^2 Y_t &= \nabla[\nabla Y_t] \\ &= \nabla[Y_t - Y_{t-1}] \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2})\end{aligned}$$

Difference and Backshift Notation

The second order difference is given by

$$\begin{aligned}\nabla^2 Y_t &= \nabla[\nabla Y_t] \\ &= \nabla[Y_t - Y_{t-1}] \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2}\end{aligned}$$

Difference and Backshift Notation

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$$\begin{aligned}\nabla^2 Y_t &= \nabla[\nabla Y_t] \\ &= \nabla[Y_t - Y_{t-1}] \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2} \\ &= (1 - 2B + B^2)Y_t\end{aligned}$$

In the next slide we will see an example of using differencing to
remove the trend

Removing Trend via Differencing

Consider a time series with a linear trend, $Y_t = \beta_0 + \beta_1 t + \eta_t$, where $\{\eta_t\}$ is stationary. First-order differencing produces a stationary series with no time-varying trend:

$$\begin{aligned}\nabla Y_t &= Y_t - Y_{t-1} \\ &= (\beta_0 + \beta_1 t + \eta_t) - (\beta_0 + \beta_1(t-1) + \eta_{t-1}) \\ &= \beta_1 + \eta_t - \eta_{t-1}\end{aligned}$$

This is the sum of a stationary series and a constant; therefore, first-order differencing removes the linear trend.

Notes on Differencing

- ▶ A polynomial trend of order q can be removed by q -th order differencing
- ▶ Applying q th-order differencing shortens the series by q observations.
- ▶ Differencing does not allow you to estimate the trend, only to remove it. *Therefore it is not appropriate if the aim of the analysis is to describe the trend*

Seasonal Component Estimation

Let's consider a situation where a time series consists of only a seasonal component (assuming the trend has been estimated/removed). In this scenario,

$$Y_t = s_t + \eta_t,$$

with $\{s_t\}$ having period d (i.e., $s_{t+jd} = s_t$ for all integers j and t), $\sum_{t=1}^d s_t = 0$ and $\mathbb{E}(\eta_t) = 0$

- ▶ Two methods to estimate $\{s_t\}$
 - ▶ Harmonic regression
 - ▶ Seasonal mean model
- ▶ A method to remove $\{s_t\} \Rightarrow$ Lag differencing

Describing Periodic/Cyclical Behavior

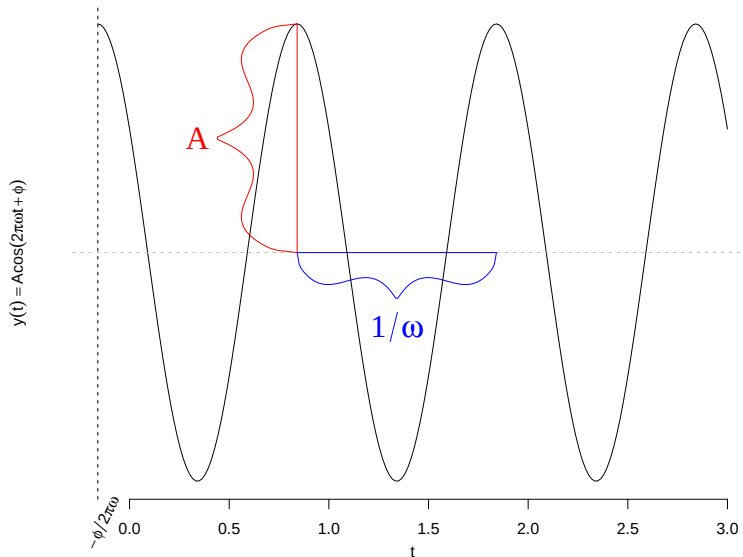
The simplest case is the **cosine wave**

$$\begin{aligned}s_t &= A \cos(2\pi\omega t + \phi) \\ &= \alpha_1 \cos(2\pi\omega t) + \alpha_2 \sin(2\pi\omega t),\end{aligned}$$

where

- ▶ A is **amplitude**
- ▶ ω is **frequency**, in cycles per time unit
- ▶ ϕ is **phase**, determining the start point of the cosine function
- ▶ $\alpha_1 = A \cos(\phi)$, $\alpha_2 = -A \sin(\phi)$, $A = \sqrt{\alpha_1^2 + \alpha_2^2}$,
 $\phi = \tan^{-1} \frac{-\alpha_2}{\alpha_1}$

Graphical Illustration of the Cosine Wave



Harmonic Regression

- ▶ A **harmonic regression** model has the form

$$s_t = \sum_{j=1}^k A_j \cos(2\pi\omega_j t + \phi_j).$$

For each $j = 1, \dots, k$:

- ▶ $A_j > 0$ is the **amplitude** of the j -th cosine wave
- ▶ ω_j controls the **frequency** of the j -th cosine wave (how often the wave repeats)
- ▶ $\phi_j \in [-\pi, \pi]$ is the **phase** of the j -th wave (where it starts)

Harmonic Regression as a Linear Model

The harmonic model can be expressed as

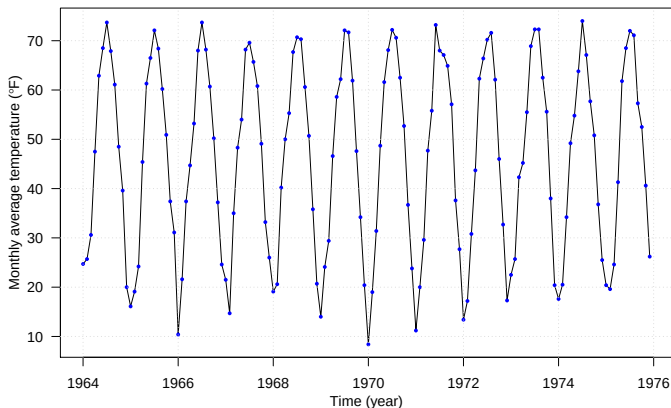
$$s_t = \sum_{j=1}^k \{ \beta_{1j} \cos(2\pi\omega_j t) + \beta_{2j} \sin(2\pi\omega_j t) \},$$

where $\beta_{1j} = A_j \cos(\phi_j)$ and $\beta_{2j} = -A_j \sin(\phi_j)$. If the frequencies are known, the coefficients can be estimated by linear regression.

The seasonal curve is nonlinear in time but linear in the unknown coefficients β_{1j} and β_{2j} .

Monthly Average Temperature in Dubuque, Iowa

Source: Cryer & Chan (2008).



Let's assume no trend in the series; our goal is to estimate s_t , the seasonal component

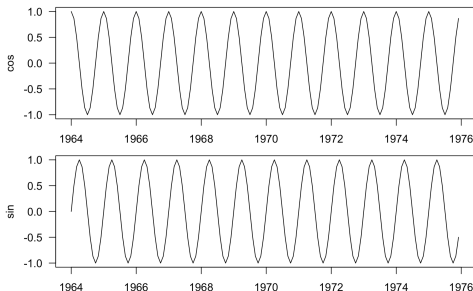
Use a Harmonic Regression to Model Annual Cycles

Model: $s_t = \beta_0 + \beta_1 \cos(2\pi t) + \beta_2 \sin(2\pi t)$.

Here t is measured in years, so the sine and cosine terms have a one-year period.

In R, we can easily create these harmonics using the `harmonic` function in the `TSA` package

```
harmonics <- harmonic(tempdub, 1)
```



R Code & Output

```
```{r}
harReg <- lm(tempdub ~ harmonics)
summary(harReg)
```

Call:

```
lm(formula = tempdub ~ harmonics)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.1580	-2.2756	-0.1457	2.3754	11.2671

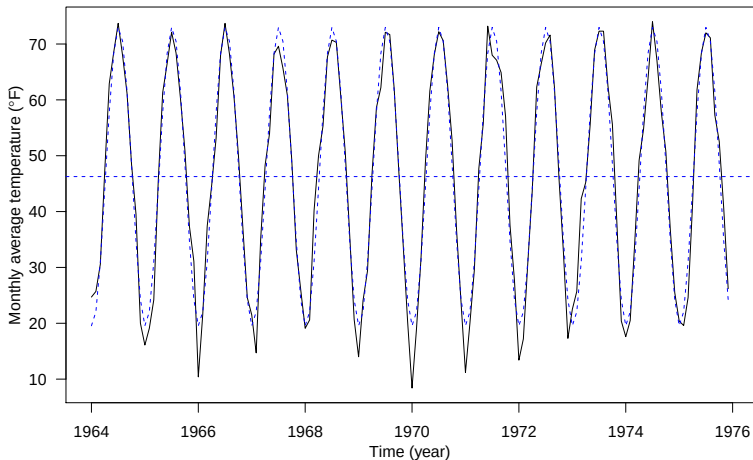
Coefficients:

	Estimate	Std. Error	t value	Pr(> t )
(Intercept)	46.2660	0.3088	149.816	< 2e-16 ***
harmonicscos(2*pi*t)	-26.7079	0.4367	-61.154	< 2e-16 ***
harmonicssin(2*pi*t)	-2.1697	0.4367	-4.968	1.93e-06 ***

---

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# The Harmonic Regression Model Fit



# Seasonal Means Model

- ▶ **Harmonics regression** assumes the seasonal pattern has a regular shape, i.e., the height of the peaks is the same as the depth of the troughs
- ▶ A less restrictive approach is to model  $\{s_t\}$  as

$$s_t = \begin{cases} \beta_1 & \text{for } t = 1, 1 + d, 1 + 2d, \dots & ; \\ \beta_2 & \text{for } t = 2, 2 + d, 2 + 2d, \dots & ; \\ \vdots & \vdots & ; \\ \beta_d & \text{for } t = d, 2d, 3d, \dots & . \end{cases}$$

- ▶ This is the **seasonal means** model, the parameters  $(\beta_1, \beta_2, \dots, \beta_d)^T$  can be estimated under the linear model framework (think about ANOVA)

# R Output

Call:

```
lm(formula = tempdub ~ month - 1)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.2750	-2.2479	0.1125	1.8896	9.8250

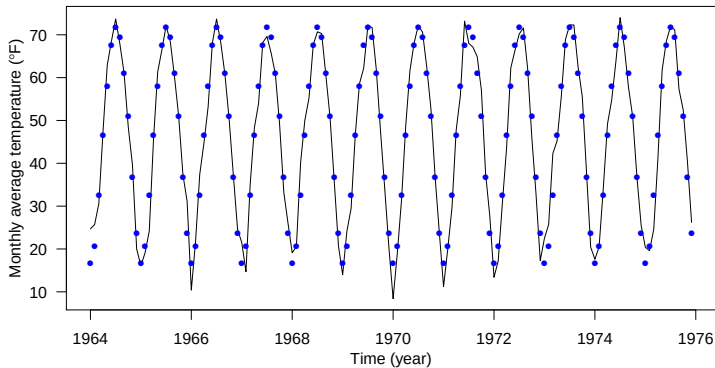
Coefficients:

	Estimate	Std. Error	t value	Pr(> t )
monthJanuary	16.608	0.987	16.83	<2e-16 ***
monthFebruary	20.650	0.987	20.92	<2e-16 ***
monthMarch	32.475	0.987	32.90	<2e-16 ***
monthApril	46.525	0.987	47.14	<2e-16 ***
monthMay	58.092	0.987	58.86	<2e-16 ***
monthJune	67.500	0.987	68.39	<2e-16 ***
monthJuly	71.717	0.987	72.66	<2e-16 ***
monthAugust	69.333	0.987	70.25	<2e-16 ***
monthSeptember	61.025	0.987	61.83	<2e-16 ***
monthOctober	50.975	0.987	51.65	<2e-16 ***
monthNovember	36.650	0.987	37.13	<2e-16 ***
monthDecember	23.642	0.987	23.95	<2e-16 ***

---

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# The Seasonal Means Model Fit



# Seasonal Differencing

- ▶ The lag- $d$  difference operator,  $\nabla_d$ , is defined by

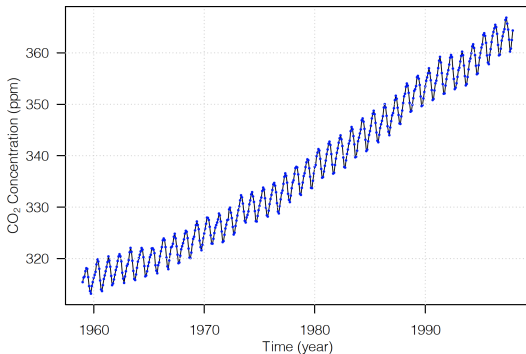
$$\nabla_d Y_t = Y_t - Y_{t-d} = (1 - B^d)Y_t.$$

Note: This is not the same as  $\nabla^d$ .

- ▶ **Example:** Suppose  $Y_t = \beta_0 + \beta_1 t + s_t + \eta_t$ , where  $s_t$  repeats every  $d$  time points and  $\eta_t$  is stationary. Lag- $d$  differencing removes the seasonal component and leaves a stationary series:

$$\begin{aligned}\nabla_d Y_t &= Y_t - Y_{t-d} \\ &= [\beta_0 + \beta_1 t + s_t + \eta_t] - [\beta_0 + \beta_1(t-d) + s_{t-d} + \eta_{t-d}] \\ &= d\beta_1 + \eta_t - \eta_{t-d}\end{aligned}$$

# Estimating the Trend and Seasonal Variation Together

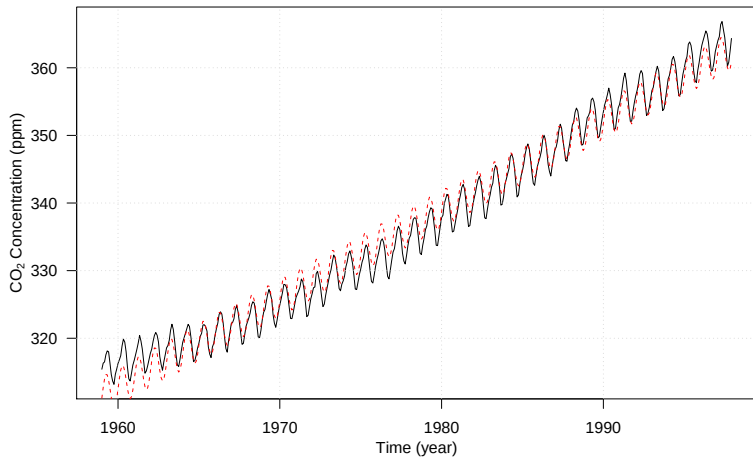


Let's perform a regression analysis to model both  $\mu_t$  (assuming a linear time trend) and  $s_t$  (using cos and sin)

```
```{r}
time <- as.numeric(time(co2))
harmonics <- harmonic(co2, 1)

lm_trendSeason <- lm(co2 ~ time + harmonics)
summary(lm_trendSeason)
```

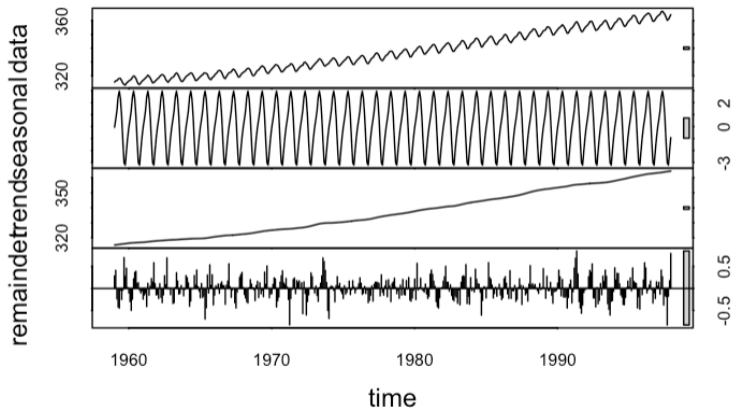

The Regression Fit



STL Decomposition

Source: Cleveland et al. (1990).

```
```{r}  
Seasonal and Trend decomposition using Loess (STL)
par(mar = c(4, 3.6, 0.8, 0.6))
stl <- stl(co2, s.window = "periodic")
plot(stl, las = 1)
```
```



Seasonal and Trend Decomposition Using LOESS (STL)

- ▶ Splits a series into **trend**, **seasonal**, and **remainder**.
- ▶ Uses **LOESS smoothing** for flexible, nonlinear patterns.
- ▶ Allows the **seasonal pattern to evolve over time** by re-estimating it locally.

Remark: Unlike parametric regression (which assumes fixed forms such as polynomials and sinusoids), STL is **nonparametric**, using local smoothing so that trend and seasonality can change flexibly over time

Summary

These slides cover:

- ▶ Main features of a time series: trend, seasonality, and “noise”
- ▶ Estimating trends using multiple linear regression (parametric) and nonparametric smoothing
- ▶ Estimating seasonality using harmonic regression (parametric) and the seasonal mean model
- ▶ Removing trends and seasonality using differencing
- ▶ Estimating trend and seasonality jointly with parametric regression models or nonparametric approaches such as STL